Parallel Mean-Comparison Simulation Settings

1 Objective

The simulation evaluates inference for the projected null

$$H_0^{\mathrm{proj}}(v_1) : (\mu_X - \mu_Z)^\top v_1 = 0, \tag{1}$$

where v_1 is the leading eigenvector of the common population covariance matrix $\Sigma = \Sigma_X = \Sigma_Z$. We compare the debiased PC, oracle PC, and plug-in PC procedures. No global-null experiment and no anchored projection procedure are included in this simulation.

We use

$$p \in \{100, 1000\}, \qquad N_X = N_Z \in \{50, 200, 400, 600, 800\}.$$

For every combination of p and the per-group sample size, we generate 1000 independent Monte Carlo repetitions. All tests are two-sided and use significance level 0.05.

2 Population covariance matrices

For each p, we first generate a sparse symmetric matrix A with zero diagonal. We sample

$$\mathrm{round}(8p/2) = 4p$$

distinct unordered pairs of coordinates uniformly at random. For every selected pair, its two symmetric entries in A are assigned the same independent Uniform$(0.5, 1.5)$ weight. Thus, each row has eight nonzero off-diagonal entries on average, but the locations of these entries do not follow a block or Toeplitz pattern.

The population covariance matrix is

$$\Sigma = I_p + 0.9 \frac{A}{\|A\|_{\mathrm{op}}}. \tag{2}$$

Since every eigenvalue of $A/\|A\|_{\mathrm{op}}$ is in $[-1, 1]$,

$$\lambda_{\min}(\Sigma) \geq 0.1.$$

We retain the first generated matrix satisfying

$$\lambda_1(\Sigma) - \lambda_2(\Sigma) \geq 0.2. \tag{3}$$

The search uses random seed 20260825 and allows at most 500 attempts. The accepted matrices used throughout the simulation have the following diagnostics.

p	Nonzero pairs	$\lambda_1(\Sigma)$	$\lambda_2(\Sigma)$	$\lambda_1(\Sigma) - \lambda_2(\Sigma)$	$\lambda_{\min}(\Sigma)$
100	400	1.900	1.553	0.347	0.445
1000	4000	1.900	1.600	0.300	0.389

The two accepted covariance matrices are fixed across all sample sizes and Monte Carlo repetitions. Therefore, changes across sample sizes reflect estimation accuracy rather than changes in the underlying covariance matrix.

3 Population mean differences

For either mean configuration described below, observations are generated as

$$X_i \overset{\text{IID}}{\sim} \mathcal{N}_p\left(\frac{\mu_X - \mu_Z}{2}, \Sigma\right), \qquad Z_i \overset{\text{IID}}{\sim} \mathcal{N}_p\left(-\frac{\mu_X - \mu_Z}{2}, \Sigma\right).$$

This symmetric parameterization makes the population mean difference exactly $\mu_X - \mu_Z$.

Let $a_1, \ldots, a_{10}$ be the indices of the ten largest entries of v_1 in absolute value, ordered from largest to smallest, and define the normalization constant

$$D = \left(\sum_{\ell=1}^{10} v_{1,a_\ell}^2\right)^{1/2}.$$

Under the projected null, the mean difference has exactly these ten active coordinates. For $k = 1, \ldots, 5$, set

$$(\mu_X - \mu_Z)_{a_{2k-1}} = \frac{v_{1,a_{2k}}}{D}, \qquad (\mu_X - \mu_Z)_{a_{2k}} = -\frac{v_{1,a_{2k-1}}}{D}, \tag{4}$$

and set all remaining coordinates to zero. In words, we divide the ten selected coordinates into five consecutive pairs. Within every pair, we swap the two leading-PC loadings and change the sign of the second one.

Each pair makes a zero contribution to the inner product with v_1:

$$(\mu_X - \mu_Z)^\top v_1 = \frac{1}{D}\sum_{k=1}^{5}\left(v_{1,a_{2k-1}}v_{1,a_{2k}} - v_{1,a_{2k}}v_{1,a_{2k-1}}\right) = 0.$$

Moreover,

$$\|\mu_X - \mu_Z\|_2^2 = \frac{1}{D^2}\sum_{k=1}^{5}\left(v_{1,a_{2k}}^2 + v_{1,a_{2k-1}}^2\right) = 1.$$

Thus, the projected-null signal has unit Euclidean norm, is supported on ten genes, and is exactly orthogonal to v_1.

For $p = 100$, the actual nonzero entries, rounded to three decimals, are

$$\begin{aligned}&\big(\,(\mu_X - \mu_Z)_{78}\,, (\mu_X - \mu_Z)_{51}\,, (\mu_X - \mu_Z)_{62}\,, (\mu_X - \mu_Z)_{86}\,, (\mu_X - \mu_Z)_{33}\,,\\ &\quad(\mu_X - \mu_Z)_{7}\,, (\mu_X - \mu_Z)_{87}\,, (\mu_X - \mu_Z)_{84}\,, (\mu_X - \mu_Z)_{99}\,, (\mu_X - \mu_Z)_{10}\,\big)\\ &\quad= (0.316, -0.442, 0.315, -0.315, 0.298, -0.305, 0.289, -0.294, 0.272, -0.285).\end{aligned}$$

For $p = 1000$, the corresponding entries are

$$\begin{aligned}&\big((\mu_X - \mu_Z)_{992}\,, (\mu_X - \mu_Z)_{338}\,, (\mu_X - \mu_Z)_{446}\,, (\mu_X - \mu_Z)_{527}\,, (\mu_X - \mu_Z)_{35}\,, \\ &\ (\mu_X - \mu_Z)_{721}\,, (\mu_X - \mu_Z)_{10}\,, (\mu_X - \mu_Z)_{20}\,, (\mu_X - \mu_Z)_{800}\,, (\mu_X - \mu_Z)_{163}\,\big) \\ &\ = (0.368, -0.379, 0.320, -0.355, 0.292, -0.298, 0.281, -0.291, 0.277, -0.279).\end{aligned}$$

Thus, the two population means are different even though (1) holds. Rejection probabilities under (4) estimate the Type I error of the projected tests.

For the power experiment, let $S = \{a_1, \dots, a_{10}\}$, and let $v_{1,S}$ agree with v_1 on S and equal zero outside S. Starting from the mean difference in (4), we add

$$0.25 \frac{v_{1,S}}{v_1^\top v_{1,S}}. \tag{5}$$

Consequently, the mean difference under the alternative satisfies

$$(\mu_X - \mu_Z)^\top v_1 = 0.25. \tag{6}$$

The addition in (5) is supported on the same ten coordinates as the projected-null mean difference. The complete mean difference under the alternative is therefore also supported on ten genes.

4 Training-fold nuisance estimators

The plug-in and debiased procedures use five-fold cross-fitting. Each group is randomly partitioned into five equal folds. For each fold $m \in [5]$, nuisance quantities are estimated using only $\mathcal{D}^{(-m)}$, and the resulting quantities are evaluated using the held out observations in $\mathcal{D}^{(m)}$. The signs of both the population and estimated leading eigenvectors are fixed by requiring their largest entry in absolute value to be positive.

4.1 Thresholded covariance estimator

Within $\mathcal{D}^{(-m)}$, the observations from the two groups are centered separately using their training-fold sample means and then pooled. Their cross-product matrix divided by $|\mathcal{D}^{(-m)}| - 2$ is denoted by $\widetilde{\Sigma}^{(-m)}$. The subtraction of two accounts for the two estimated group means.

The diagonal entries are retained. For $j \neq j'$, the covariance estimator is

$$\Sigma^{(-m)}_{jj'} = \widetilde{\Sigma}^{(-m)}_{jj'} \mathbf{1}\left\{ \left| \widetilde{\Sigma}^{(-m)}_{jj'} \right| \geq 1.25 \left[\left\{ \widetilde{\Sigma}^{(-m)}_{jj} \widetilde{\Sigma}^{(-m)}_{j'j'} + \left(\widetilde{\Sigma}^{(-m)}_{jj'} \right)^2 \right\} \frac{\log p}{|\mathcal{D}^{(-m)}| - 2} \right]^{1/2} \right\}. \tag{7}$$

If the smallest eigenvalue of the matrix in (7) is less than 10^{-6}, the minimum multiple of I_p needed to make its smallest eigenvalue equal to 10^{-6} is added. This diagonal shift changes neither the eigenvectors nor the eigenvalue gaps. The leading eigenvalue and eigenvector of the resulting matrix are denoted by $\lambda_1^{(-m)}$ and $v_1^{(-m)}$.

This estimator is an entrywise-thresholded covariance estimator. In particular, the revision simulation does not use `PMA::SPC` or the spiked covariance estimator used elsewhere in the manuscript code.

4.2 Mean-difference estimator used in the correction

The training-fold sample means are denoted by $\mu_X^{(-m)}$ and $\mu_Z^{(-m)}$. Before constructing the one-step correction, their difference is hard-thresholded coordinatewise. For coordinate j, it is retained only when

$$\left|\mu_{X,j}^{(-m)} - \mu_{Z,j}^{(-m)}\right| \geq 1.25\sqrt{2\log p}\left(\frac{\widehat{\mathrm{Var}}^{(-m)}(X_j)}{4N_X/5} + \frac{\widehat{\mathrm{Var}}^{(-m)}(Z_j)}{4N_Z/5}\right)^{1/2}. \tag{8}$$

Here $\widehat{\mathrm{Var}}^{(-m)}(X_j)$ and $\widehat{\mathrm{Var}}^{(-m)}(Z_j)$ are the ordinary sample variances calculated from the two training-fold samples. The hard-thresholded difference in (8), rather than the unthresholded difference, is used in

$$s^{(-m)} = \left(\lambda_1^{(-m)} I_p - \Sigma^{(-m)}\right)^{+}\left(\mu_X^{(-m)} - \mu_Z^{(-m)}\right). \tag{9}$$

The unthresholded training-fold sample means are still used to center the observations appearing in the influence functions.

5 Compared methods

5.1 Oracle PC

The oracle procedure treats the population v_1 as known. It estimates

$$\theta = (\mu_X - \mu_Z)^\top v_1$$

by

$$\widehat{\theta}_{\mathrm{or}} = \left(\overline{X} - \overline{Z}\right)^\top v_1.$$

Its squared standard error is the sum of the sample variance of $X_i^\top v_1$ divided by N_X and the sample variance of $Z_i^\top v_1$ divided by N_Z. More explicitly,

$$\begin{aligned}\widehat{\sigma}_{\mathrm{or}}^2 = {} & \frac{1}{N_X(N_X-1)}\sum_{i=1}^{N_X}\left\{(X_i - \overline{X})^\top v_1\right\}^2 \\ & + \frac{1}{N_Z(N_Z-1)}\sum_{i=1}^{N_Z}\left\{(Z_i - \overline{Z})^\top v_1\right\}^2.\end{aligned} \tag{10}$$

Thus, $\widehat{\sigma}_{\mathrm{or}}$ is the estimated standard deviation of $\widehat{\theta}_{\mathrm{or}}$, not an additional model parameter. Since v_1 is an eigenvector of Σ, the population variance being estimated is

$$\mathrm{Var}(\widehat{\theta}_{\mathrm{or}}) = \lambda_1(\Sigma)\left(\frac{1}{N_X} + \frac{1}{N_Z}\right).$$

The oracle statistic is $T_{\mathrm{or}} = \widehat{\theta}_{\mathrm{or}}/\widehat{\sigma}_{\mathrm{or}}$.

5.2 Plug-in PC

For each held-out fold, the plug-in estimate is

$$\widehat{\theta}_{\mathrm{pi}}^{(m)} = \left(\mu_X^{(m)} - \mu_Z^{(m)}\right)^\top v_1^{(-m)}. \tag{11}$$

The five fold estimates are averaged. If $\widehat{\sigma}_{\text{pi}}^{2(m)}$ is the sum of the two held-out sample variances of the projected observations divided by their respective fold sample sizes, the implementation uses

$$\widehat{\theta}_{\text{pi}} = \frac{1}{5}\sum_{m=1}^{5} \widehat{\theta}_{\text{pi}}^{(m)}, \qquad \widehat{\sigma}_{\text{pi}}^{2} = \frac{1}{5^2}\sum_{m=1}^{5} \widehat{\sigma}_{\text{pi}}^{2(m)}, \qquad T_{\text{pi}} = \frac{\widehat{\theta}_{\text{pi}}}{\widehat{\sigma}_{\text{pi}}}. \tag{12}$$

5.3 Debiased PC

Using (9), the estimated influence functions in fold m are

$$\phi_X^{(-m)}(X) = s^{(-m)\top}\left[\left(X - \mu_X^{(-m)}\right)\left(X - \mu_X^{(-m)}\right)^{\top} - \Sigma^{(-m)}\right] v_1^{(-m)}, \tag{13}$$

$$\phi_Z^{(-m)}(Z) = s^{(-m)\top}\left[\left(Z - \mu_Z^{(-m)}\right)\left(Z - \mu_Z^{(-m)}\right)^{\top} - \Sigma^{(-m)}\right] v_1^{(-m)}. \tag{14}$$

Because the held-out group sizes are equal, the fold-specific one-step estimate implemented in the simulation is

$$\widehat{\theta}_{\text{1s}}^{(m)} = \widehat{\theta}_{\text{pi}}^{(m)} + \frac{1}{|\mathcal{D}^{(m)}|}\left\{\sum_{X_i\in\mathcal{D}^{(m)}} \phi_X^{(-m)}(X_i) + \sum_{Z_i\in\mathcal{D}^{(m)}} \phi_Z^{(-m)}(Z_i)\right\}. \tag{15}$$

For variance estimation, the corresponding held-out scores are

$$\begin{aligned} V_X^{(-m)}(X) &= X^{\top} v_1^{(-m)} + \frac{1}{2}\phi_X^{(-m)}(X), \\ V_Z^{(-m)}(Z) &= Z^{\top} v_1^{(-m)} - \frac{1}{2}\phi_Z^{(-m)}(Z). \end{aligned}$$

The fold variance is the sum of the two sample variances of these scores divided by their respective held-out sample sizes. The fold estimates and variances are then combined exactly as in (12), giving $\widehat{\theta}_{\text{1s}}$, $\widehat{\sigma}_{\text{1s}}$, and

$$T_{\text{1s}} = \frac{\widehat{\theta}_{\text{1s}}}{\widehat{\sigma}_{\text{1s}}}.$$

For all three methods, the two-sided p-value is

$$2\Phi\left(-|T|\right),$$

where Φ is the standard normal distribution function. Rejection occurs when this value is at most 0.05.

6 Monte Carlo summaries and parallel execution

For each combination of p, per-group sample size, method, and simulation target, the reported rejection rate is the average of the 1000 rejection indicators. If the rejection rate is $\widehat{q}$, its reported Monte Carlo standard error is

$$\left\{\frac{\widehat{q}(1-\widehat{q})}{1000}\right\}^{1/2}.$$

Under (4), this rejection rate estimates Type I error. Under (6), it estimates power.

For computation, ten consecutive repetitions for one (p, N_X) combination form a single parallel task. Hence, the complete experiment contains

$$2 \times 5 \times 100 = 1000$$

parallel tasks. A task has a self-describing identifier such as `p1000_n0200_b037`. Each task writes to its own file, so completed tasks can be validated and skipped when the pipeline is resumed. The random seed for repetition r is

$$20260829 + p \times 10^6 + N_X \times 10^3 + r.$$

The fold assignments use this seed plus one under the projected null and this seed plus two under the projected alternative. The parallel execution order therefore does not affect the simulated data or the cross-fitting splits.

The settings are declared in `config/011_mean_comparison_simulation_config.R`. Population quantities and the task manifest are prepared by `code/011_prepare_mean_comparison_tasks.R`; a single task is run by `code/012_run_mean_comparison_task.R`; and the computational interface is contained in `src/012_mean_comparison_simulation_task.R`. The parallel dispatcher, result collector, complete runner, and plotting script are numbered 013 through 016, respectively.